Software Requirements Specification (SRS)

Peer-to-Peer Tutoring Scheduler

Project	Peer-to-Peer Tutoring Scheduler
Version	1.0
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Revision history

Version	Date	Author	Change summary
1.0	27-10-2026	Team	SRS with diagrams, RTM and quality attributes

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

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1. Introduction

1.1 Purpose

This SRS defines the functional and non-functional requirements of the Peer-to-Peer Tutoring Scheduler, a web booking system in which students publish tutoring slots and peers browse availability to reserve one-on-one sessions. It is the reference for design (SAD), testing (STP) and the GitHub project backlog used in Sprint 1 and Sprint 2.

1.2 Scope

In scope: user registration and login, tutor profiles and subjects, publishing (single and recurring) availability slots, a calendar UI for browsing and booking, booking conflict resolution, cancellation and rescheduling, email confirmation and reminder automation through background jobs, ratings and reviews, and basic admin moderation.

Out of scope: video conferencing (sessions take place on the tutor's own meeting link or in person), payments, and native mobile apps.

1.3 Audience

Developers, QA engineers, the course instructor and evaluators, and future maintainers.

1.4 Definitions and acronyms

Term	Meaning
Tutor	A student who offers tutoring slots
Learner	A student who books a slot
Slot	A time window published by a tutor (start, end, subject, mode)
Booking	A reservation of one slot by one learner
Conflict	Two bookings overlapping for the same tutor or the same learner
Job / Worker	Background task processed asynchronously from a queue
JWT, API, UI, RBAC, TLS, CI/CD, RTM	Standard software engineering acronyms

2. Overall description

2.1 Product perspective

A standalone three-tier web application: a Next.js (React) web frontend with a calendar component, a Node.js/Express REST API, a PostgreSQL database, and a Redis-backed job queue whose worker sends emails through an SMTP provider.

2.2 Major product functions

- Register, verify email, log in and manage a profile
- Publish, edit and delete availability slots (single or weekly recurring)
- Browse and filter slots by subject, tutor, date and time in a calendar view
- Reserve a slot with automatic conflict detection
- Cancel or reschedule a booking within the allowed window
- Automatic confirmation, 24-hour and 1-hour reminder emails
- Rate and review completed sessions
- Admin management of users and reviews

2.3 User roles

Role	Characteristics
Learner	Any registered student; wants to find help quickly on a phone or laptop
Tutor	Registered student who has enabled tutoring and listed subjects
Admin	Course or platform staff who moderate accounts and content

2.4 Operating environment

Modern browsers (Chrome, Firefox, Edge, Safari, latest two versions) on desktop and mobile. Server runs in Docker containers on Linux (Node.js 20 LTS, PostgreSQL 16, Redis 7).

2.5 Constraints

- **Data protection:** student personal data (names, emails, schedules) must be handled according to the principles of India's Digital Personal Data Protection (DPDP) Act 2023: collect only what is needed, and never expose emails publicly.
- **Security standards:** all communication must use TLS 1.2 or higher; passwords must be stored only as salted hashes (bcrypt), following OWASP guidance.
- **Email delivery:** reminders depend on an external SMTP / email API provider and its sending limits and spam policies.
- **Time handling:** all times must be stored in UTC and converted to the user's local time zone for display.
- **Client platform:** the application must run in standard web browsers (latest two versions of Chrome, Firefox, Edge and Safari) without any plugin or installation.
- **Technology:** the system must be built on an open-source stack (Next.js, Node.js/Express, PostgreSQL, Redis) and be deployable as Docker containers.

2.6 Assumptions

Students use their institutional email; an SMTP provider sandbox (e.g. Mailtrap or SendGrid free tier) is available; all times are stored in UTC and shown in the user's local time zone.

3. External interface requirements

3.1 User interfaces

Responsive web UI with: login/register pages, a calendar page (month, week and day views), a slot-creation dialog, a booking confirmation dialog, “My bookings” list, tutor profile page and admin dashboard. Colour-blind-safe status colours (free, booked, mine) with text labels as well.

3.2 Hardware interfaces

None beyond standard client devices and a cloud/VM host.

3.3 Software interfaces

- REST API (JSON over HTTPS) between frontend and backend
- PostgreSQL via an ORM (Prisma)
- Redis + BullMQ for delayed jobs
- SMTP / SendGrid API for email
- Optional iCalendar (.ics) export

3.4 Communications

HTTPS (TLS 1.2+) for all client traffic; email via authenticated SMTP over TLS.

4. System features (functional requirements)

IDs follow **P2P-F-###**. Priority: High (H), Medium (M), Low (L).

4.1 Account and authentication

Req ID	Requirement (shall…)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
P2P-F-001	The system shall allow a student to register with name, institutional email and password.	Functional	H	Students	Account created; duplicate email rejected with a clear message. Test: TC-AUTH-01	Requires email service
P2P-F-002	The system shall verify the email address through a time-limited link before booking is allowed.	Functional	M	Security	Unverified users cannot book; link expires after 24 h. Test: TC-AUTH-02	Depends on Notification Service
P2P-F-003	The system shall authenticate users with email and password and issue a JWT session.	Functional	H	Security	Valid login returns token; invalid login returns 401. Test: TC-AUTH-03	Depends on F-001
P2P-F-004	The system shall let a user enable the Tutor role and list subjects and a short bio.	Functional	H	Tutors	Profile shows subjects; tutor appears in search. Test: TC-PROF-01	Depends on F-003

4.2 Availability slots (tutor)

Req ID	Requirement (shall…)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
P2P-F-005	The system shall let a tutor create a slot with date, start time, end time, subject and mode (online link or location).	Functional	H	Tutors	Slot saved and visible on calendar. Test: TC-SLOT-01	Depends on F-004
P2P-F-006	The system shall let a tutor create weekly recurring slots for up to 8 weeks.	Functional	M	Tutors	Correct number of slot instances generated. Test: TC-SLOT-02	Builds on F-005
P2P-F-007	The system shall prevent a tutor from creating overlapping slots.	Functional	H	Tutors / Business rule	Overlapping slot rejected with 409 and details. Test: TC-SLOT-03	Validated in Slot Service
P2P-F-008	The system shall let a tutor edit or delete an unbooked slot; deleting a booked slot cancels the booking and notifies the learner.	Functional	M	Tutors / Learners	Learner gets cancellation email. Test: TC-SLOT-04	Triggers F-021

4.3 Browsing and calendar UI

Req ID	Requirement (shall…)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
P2P-F-009	The system shall display available slots in month, week and day calendar views.	Functional	H	Learners	Slots render in correct local time. Test: TC-CAL-01	Calendar component, time-zone conversion
P2P-F-010	The system shall let learners filter slots by subject, tutor name, date range and mode.	Functional	H	Learners	Only matching slots shown. Test: TC-CAL-02	Needs DB indexes on subject, start time
P2P-F-011	The system shall show a learner's own bookings on the same calendar in a distinct style.	Functional	M	Learners	"Mine" style and label visible. Test: TC-CAL-03	Depends on F-012

4.4 Booking and conflict resolution

Req ID	Requirement (shall…)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
P2P-F-012	The system shall let a learner reserve a free slot for a one-on-one session.	Functional	H	Learners	Booking created; slot status becomes BOOKED. Test: TC-BOOK-01	Depends on F-002, F-009
P2P-F-013	The system shall guarantee that a slot can be booked by only one learner even under concurrent requests.	Functional	H	Business rule	Of N simultaneous requests exactly 1 succeeds, others get 409. Test: TC-BOOK-02	DB transaction + row lock
P2P-F-014	The system shall reject a booking that overlaps another booking of the same learner.	Functional	H	Business rule	409 returned with the clashing booking. Test: TC-BOOK-03	Overlap query in Booking Service
P2P-F-015	When a conflict occurs, the system shall suggest up to 3 nearest free slots for the same subject.	Functional	M	Learners	Suggestions returned in response and shown in UI. Test: TC-BOOK-04	Depends on F-013, F-014
P2P-F-016	The system shall let a learner cancel a booking up to 2 hours before start, freeing the slot.	Functional	H	Learners / Tutors	Slot returns to FREE; both parties notified. Test: TC-BOOK-05	Triggers F-021
P2P-F-017	The system shall let a learner reschedule by atomically moving the booking to another free slot.	Functional	M	Learners	Old slot freed and new slot booked in one transaction. Test: TC-BOOK-06	Atomic transaction; triggers F-021
P2P-F-018	The system shall block learners from booking their own slots.	Functional	M	Business rule	Attempt returns 403. Test: TC-BOOK-07	Ownership check

4.5 Notifications and background jobs

Req ID	Requirement (shall…)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
P2P-F-019	The system shall send a booking confirmation email to tutor and learner within 1 minute of booking.	Functional	H	Learners / Tutors	Email received in sandbox inbox. Test: TC-NOTIF-01	Redis queue + SMTP provider
P2P-F-020	The system shall schedule background reminder jobs 24 hours and 1 hour before each session.	Functional	H	Tutors (fewer no-shows)	Jobs exist in queue with correct delay. Test: TC-NOTIF-02	Redis queue + worker
P2P-F-021	The system shall skip reminders for cancelled bookings and reschedule them for moved bookings.	Functional	H	Learners / Tutors	No reminder sent after cancellation. Test: TC-NOTIF-03	Depends on F-016, F-017
P2P-F-022	The system shall retry failed email jobs up to 3 times with exponential backoff and log final failure.	Functional	M	Operations	Retry count and FAILED status logged. Test: TC-NOTIF-04	Worker retry policy

4.6 Reviews and administration

Req ID	Requirement (shall…)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
P2P-F-023	The system shall let a learner rate (1 to 5) and review a session after it ends, once per booking.	Functional	M	Learners	Average rating updates on tutor profile. Test: TC-REV-01	Only after session end time
P2P-F-024	The system shall let an admin suspend users and hide abusive reviews.	Functional	L	Admin	Suspended users cannot log in. Test: TC-ADM-01	RBAC (SR-003)

5. Non-functional requirements

Req ID	Requirement	Category	Priority	Acceptance / measurement		
P2P-NF-001	95% of API requests shall respond within 500 ms at 100 concurrent users.	Performance	H	k6/JMeter load test. TC-PERF-01		
P2P-NF-002	The calendar page shall load (first contentful paint) in under 2 s on a 4G connection.	Performance	M	Lighthouse report. TC-PERF-02		
P2P-NF-003	The service shall target 99% monthly availability; reminder jobs shall survive server restarts (persistent queue).	Reliability	H	Restart test; no lost jobs. TC-REL-01		
P2P-NF-004	The UI shall be responsive from 360 px to 1920 px and meet WCAG 2.1 AA for contrast and keyboard navigation.	Usability	M	axe audit, manual check. TC-UX-01		
P2P-NF-005	A new user shall complete a first booking in 3 clicks or fewer from the calendar page.	Usability	M	Usability walkthrough. TC-UX-02		
P2P-NF-006	Backend code shall keep at least 70% unit-test coverage and pass ESLint with zero errors on every PR.	Maintainability	H	CI report. TC-CI-01		
P2P-NF-007	The whole system shall run with a single `docker compose up` on any OS with Docker.	Portability	H	Run on Windows, macOS, Linux. TC-PORT-01		

5.1 Security

5.1.1 Security objectives

1. **Confidentiality:** protect students' personal data (email, schedule) from unauthorised access.
2. **Integrity:** ensure bookings can be created, changed or cancelled only by their rightful owners, and that no double-booking can be forced.
3. **Availability:** resist abuse (spam bookings, brute-force login) that could deny service to genuine users.

5.1.2 Security requirements

Req ID	Requirement (shall…)	Type	Priority	Acceptance / test
P2P-SR-001	All traffic shall use HTTPS (TLS 1.2+); cookies shall be Secure and HttpOnly.	Security	H	TLS scan. TC-SEC-01
P2P-SR-002	Passwords shall be hashed with bcrypt (cost 12 or higher) and never logged.	Security	H	DB and log inspection. TC-SEC-02
P2P-SR-003	Every API endpoint shall enforce role-based access control (Learner, Tutor, Admin) and ownership checks.	Security	H	Unauthorised access returns 403. TC-SEC-03
P2P-SR-004	Login and booking endpoints shall be rate-limited (e.g. 5 failed logins per 15 min, 20 bookings per hour).	Security	M	Excess requests return 429. TC-SEC-04
P2P-SR-005	All inputs shall be validated server-side and queries parameterised to prevent SQL injection and XSS.	Security	H	OWASP ZAP scan, no high findings. TC-SEC-05
P2P-SR-006	Secrets (DB password, SMTP key, JWT secret) shall be stored in environment variables / GitHub Secrets, never in the repo.	Security	H	Secret scan in CI. TC-SEC-06

6. Quality attributes and acceptance tests

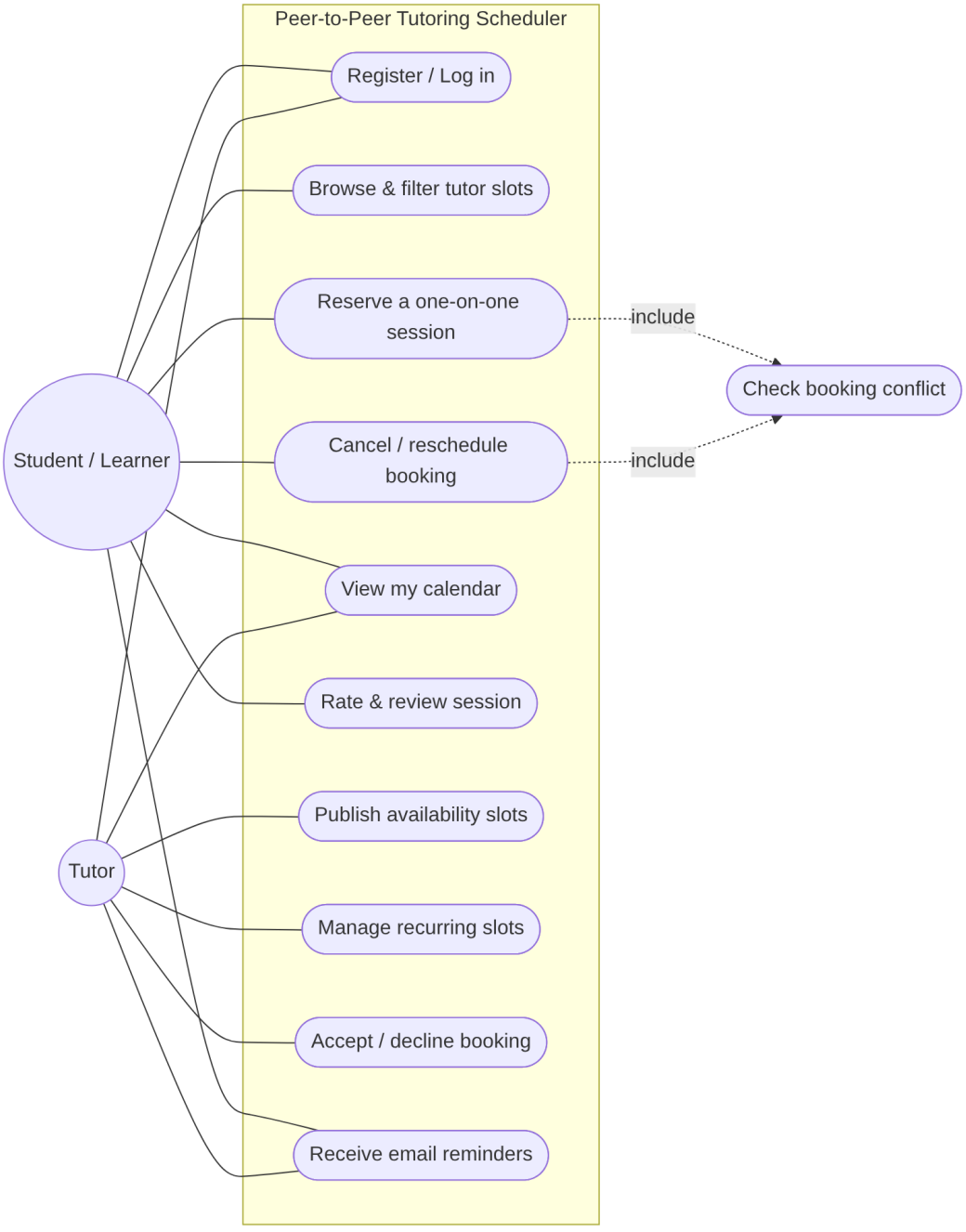
Attribute	Does the system achieve it? (summary)
Functionality	Yes. All 24 FRs map to test cases in the RTM; the core flow (publish slot, browse, book, remind) is verified end to end, and conflict handling is proven by a concurrency test.
Reliability	Database transactions with row locks prevent double booking; the Redis queue is persistent so reminders survive restarts; failed emails are retried 3 times.
Usability	Familiar calendar views, colour plus text labels, 3-click booking, helpful conflict messages with alternative slots, responsive and WCAG 2.1 AA.
Efficiency	Indexed queries on (tutor_id, start_time), paginated date-range fetching, and email sending moved off the request path into background jobs keep APIs under 500 ms.
Maintainability	Layered, modular services, consistent linting, 70%+ test coverage, CI on every PR and documented API (OpenAPI).
Portability	Dockerised services, environment-based configuration, browser-based client; runs on any OS and can deploy to any cloud (Render, Railway, AWS).

Exit criteria: all High-priority FRs implemented and verified, no open critical defects, NFR targets met, and RTM shows all tests passed.

Acceptance suites: Authentication, Slots, Calendar, Booking and Conflicts, Notifications, Reviews, Performance, Security, Usability and Portability.

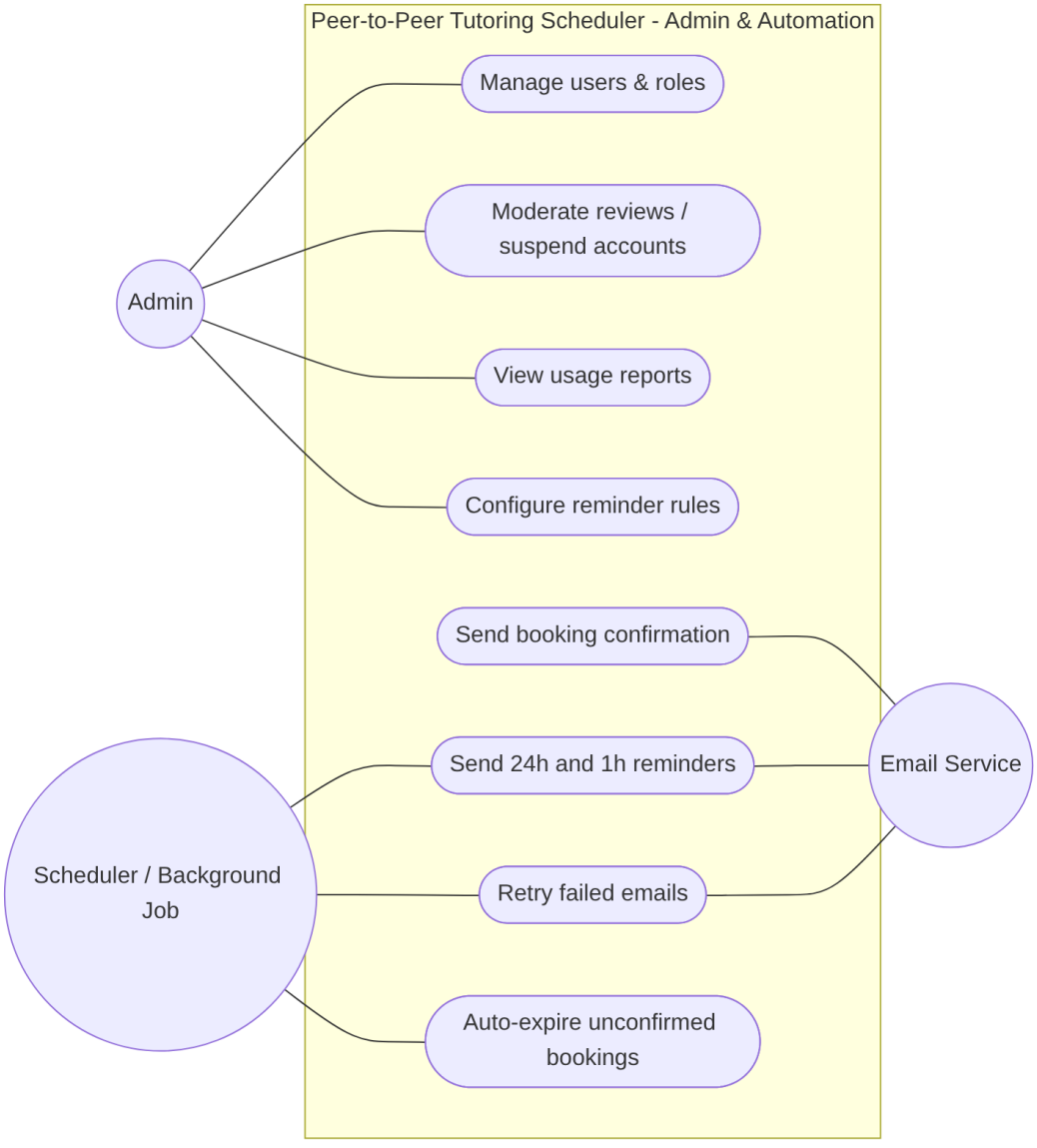
7. UML use-case diagrams

7.1 Use-case diagram 1: Learner and Tutor



Use-case diagram for learners and tutors

7.2 Use-case diagram 2: Admin and background automation



Use-case diagram for admin and background jobs

8. Requirements Traceability Matrix (RTM)

Req ID	Short name	Design ref (SAD)	Module	Test case(s)	Status (N/P/A)	Comments
P2P-F-001	Register	3.4 / API 4.3	Auth Service	TC-AUTH-01	N	
P2P-F-003	Login	3.4	Auth Service	TC-AUTH-03	N	
P2P-F-005	Create slot	3.4 / API 4.3	Slot Service	TC-SLOT-01	N	
P2P-F-007	No overlapping slots	3.4	Slot Service	TC-SLOT-03	N	
P2P-F-009	Calendar views	4.5	Calendar UI	TC-CAL-01	N	
P2P-F-012	Reserve slot	4.2 Seq-1	Booking Service	TC-BOOK-01	N	
P2P-F-013	Concurrency-safe booking	4.2 Seq-1	Booking Service	TC-BOOK-02	N	
P2P-F-014	Learner overlap check	4.2 Seq-1	Booking Service	TC-BOOK-03	N	
P2P-F-016	Cancel booking	3.4	Booking Service	TC-BOOK-05	N	
P2P-F-019	Confirmation email	4.2 Seq-2	Notification Service	TC-NOTIF-01	N	
P2P-F-020	Reminder jobs	4.2 Seq-2	Reminder Worker	TC-NOTIF-02	N	
P2P-F-022	Email retry	4.2 Seq-2	Reminder Worker	TC-NOTIF-04	N	
P2P-F-023	Reviews	3.4	Review Service	TC-REV-01	N	
P2P-NF-001	API response time	3.6	API / DB	TC-PERF-01	N	
P2P-NF-003	Reliability of jobs	3.7	Job Queue	TC-REL-01	N	
P2P-NF-007	Portability	3.6	Docker	TC-PORT-01	N	
P2P-SR-002	Password hashing	3.9	Auth Service	TC-SEC-02	N	
P2P-SR-003	RBAC	3.9	API Gateway	TC-SEC-03	N	

Status key: N = Not tested, P = Passed, A = Accepted. Remaining requirements follow the same mapping and are tracked as backlog items in GitHub Projects.